Kinetic energy, spaceship, relative speed, explosion.

10 A spaceship of mass M is at rest. It separates into two parts in an explosion in which the total kinetic energy released is E. Immediately after the explosion the two parts have masses m_1 and m_2 and speeds v_1 and v_2 respectively. Show that the minimum possible relative speed $v_1 + v_2$ of the two parts of the spaceship after the explosion is $(8E/M)^{1/2}$.